

Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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Executive Summary

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Time-Series Forecasting

Time-series forecasting stands as a cornerstone of modern predictive analytics, vital for interpreting and anticipating the dynamic, state-dependent behaviors inherent to complex systems. Whether predicting fluctuating loads in renewable energy grids, identifying macro-economic trends, or monitoring structural health in industrial machinery, the ability to project historical sequential data into future horizons shifts

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During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.1 Pro, Gemini 3.5 Flash, Sonnet 4.6, and Opus 4.7, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

organizational strategy from reactive mitigation to proactive optimization. As systems grow increasingly interconnected and generate data at unprecedented scales, the demand for forecasting frameworks capable of parsing intricate temporal dynamics has driven a massive evolution in modeling methodologies.

Signal Representation and Aliasing Vulnerabilities

Regardless of the downstream modeling architecture utilized, the predictive fidelity of any forecasting pipeline remains strictly bounded by the digital representation of the underlying signal. This representation relies heavily on initial data acquisition, signal conditioning, and pre-processing protocols.

The fundamental mathematical boundary governing the digitization of continuous physical processes is the Nyquist-Shannon sampling theorem. The theorem dictates that a continuous-time signal containing frequency components can be perfectly captured and reconstructed without loss of information if and only if it is sampled uniformly at a sampling frequency f_s that is strictly greater than twice the highest frequency component ($f_{\max}$) present within the signal:

$$f_s > 2f_{\max}. \quad (1)$$

The threshold $f_N = \frac{f_s}{2}$ is defined as the Nyquist frequency. When a system violates this criterion, either due to an insufficient sampling rate or the absence of a proper low-pass anti-aliasing filter, high-frequency components beyond the Nyquist limit fold back into the lower spectrum. This distortive phenomenon, known as *aliasing*, introduces spurious, artificial low-frequency components that permanently mask and corrupt the true underlying dynamics of the signal.

Evolution of Forecasting Methodologies

Historically, time-series analysis relied on classical statistical models such as AutoRegressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS). These traditional methods are deeply rooted in mathematical theory, offering high interpretability, exact statistical guarantees, and robust performance on stable, low-dimensional datasets. However, they rely heavily on strict underlying assumptions such as linearity, and stationarity.

To capture complex, non-linear temporal behaviors without rigid feature engineering, the paradigm shifted toward deep learning. Early architectures leveraged Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Convolutional Neural Networks (CNNs) to map temporal sequences sequentially.

More recently, the state of the art has been dominated by transformer-based architectures. By replacing recurrent bottlenecks with self-attention mechanisms, transformers can process sequences in parallel and explicitly compute dependencies between distant time steps. This allows the model to learn both localized high-frequency fluctuations and long-range macro-patterns simultaneously, forming highly adaptive architectures that can be fine-tuned to specific downstream industrial or scientific applications.

Representing the cutting edge of this evolution is the emergence of time-series foundation models, most notably exemplified by the *Chronos* family of frameworks. Unifying time-series forecasting with the architectures powering modern natural language processing, Chronos treats time-series values not as continuous regressions, but as discrete language tokens.

The framework tokenizes real-valued historical data by quantizing values into discrete buckets via a scaling and binning operator. These numerical tokens are then fed directly into standard transformer language model backbones (such as T5). Trained on billions of diverse data points spanning synthetic and real-world domains, Chronos exhibits powerful zero-shot generalization, accurately predicting future trajectories across entirely unseen systems without requiring task-specific structural retraining or structural fine-tuning. This reliance on discrete tokenization windows, however, introduces novel architectural interactions with signal frequencies, exposing the model to localized structural vulnerabilities when processing periodic inputs.

Patch-Based Tokenization: a Formal Description

Let $\mathbf{x} = \{x[n]\}_{n \in \mathbb{Z}}$ be a discrete-time signal representing the input time series. We define the patch-based tokenization operator $\mathcal{T}_{P,S}$ with patch length $P \in \mathbb{N}^+$ and stride $S \in \mathbb{N}^+$ as a mapping that transforms the signal $\mathbf{x}$ into a sequence of vector-valued tokens $\mathbf{T} = \{\mathbf{t}_k\}_{k \in \mathbb{Z}}$, where each token $\mathbf{t}_k \in \mathbb{R}^P$.

The k -th token is defined as

$$\mathbf{t}_k = \mathcal{T}_{P,S}(\mathbf{x})[k] = \begin{bmatrix} x[kS] \\ x[kS + 1] \\ \vdots \\ x[kS + P - 1] \end{bmatrix}. \quad (2)$$

Using element-wise indexing notation,

$$\mathbf{t}_k[m] = x[kS + m], \quad m \in \{0, \dots, P - 1\}. \quad (3)$$

Phase-Locking and Patch Indistinguishability

Consider a continuous-time periodic signal sampled uniformly at sampling frequency f_s , yielding the discrete-time sequence $x[n]$. We isolate a fundamental harmonic component and model it as

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \quad (4)$$

where f denotes the signal frequency satisfying the Nyquist criterion $f < \frac{f_s}{2}$, and $\phi \in [0, 2\pi)$ is an arbitrary phase offset.

Phase-locking occurs when the structural phase profile of the signal repeats exactly under a translational shift equal to the stride S . Evaluating a stride translation yields

$$x[n + S] = \sin\left(2\pi \frac{f}{f_s} (n + S) + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + 2\pi \frac{fS}{f_s} + \phi\right). \quad (5)$$

For exact alignment between the signal and the tokenization operator, the phase accumulation over one stride must correspond to an integer number of 2π -periods

$$2\pi \frac{fS}{f_s} = 2\pi c, \quad c \in \mathbb{N}^+. \quad (6)$$

As a result, the signal exhibits exact stride-period translational invariance

$$x[n + S] = \sin\left(2\pi \frac{f}{f_s} n + 2\pi c + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + \phi\right) = x[n]. \quad (7)$$

Solving Equation (6) for f yields the phase-locking frequency class, which is the class of frequencies that produce exact stride-period translational invariance, we denote that as $\mathcal{F}_{\text{lock}}$:

$$\mathcal{F}_{\text{lock}} = \left\{ f \in \mathbb{R}^+ \mid f = c \frac{f_s}{S}, \ c \in \mathbb{N}^+, \ f < \frac{f_s}{2} \right\}. \quad (8)$$

The collapse of consecutive tokens is not restricted to sinusoidal signals. Consider an arbitrary discrete-time signal satisfying the translational invariance condition

$$x[n + S] = x[n], \quad \forall n. \quad (9)$$

Taking into account two consecutive tokens, using the element-wise notation from Equation (3) and applying Equation (9) gives

$$\mathbf{t}_{k+1}[m] = x[(k+1)S + m] = x[kS + m + S] = x[kS + m] = \mathbf{t}_k[m]. \quad (10)$$

Since this equality holds for every component,

$$\mathbf{t}_{k+1} = \mathbf{t}_k. \quad (11)$$

Consequently, any signal exhibiting exact stride-period translational invariance produces identical consecutive tokens regardless of the patch length P .

Characterization of the Loss of Observability

When consecutive patches become indistinguishable, a localized structural aliasing event occurs. This degradation can be characterized through three complementary perspectives.

1. Rank Deficiency in the Embedding Space

Let

$$\mathbf{M} = [\mathbf{t}_k, \mathbf{t}_{k+1}, \dots, \mathbf{t}_{k+L-1}] \in \mathbb{R}^{P \times L} \quad (12)$$

be the local trajectory matrix.

If two consecutive tokens become identical, then the corresponding columns become linearly dependent. Consequently, the trajectory matrix cannot achieve full rank:

$$\text{rank}(\mathbf{M}) < \min(P, L). \quad (13)$$

More generally, every duplicated token introduces at least one linear dependency and therefore cannot increase the matrix rank. Repeated patch collapse systematically compresses the effective dimensionality of the local trajectory manifold and reduces the geometric diversity available to downstream embedding layers.

2. Content Degeneration in Self-Attention

If consecutive patches satisfy

$$\mathbf{t}_k = \mathbf{t}_{k+1}, \quad (14)$$

their content representations become identical.

For transformer architectures without positional information,

$$\mathbf{q}_k = \mathbf{q}_{k+1}, \quad \mathbf{k}_k = \mathbf{k}_{k+1}. \quad (15)$$

Consequently, rows k and $k+1$ of the content-derived attention score matrix become identical.

Modern transformer architectures generally employ positional encodings, rotary embeddings, or relative positional biases. Such mechanisms can still distinguish consecutive positions. Nevertheless, the content-derived component of the attention mechanism becomes degenerate, forcing the model to rely increasingly on positional information to break the symmetry introduced by token collapse.

3. Loss of Sub-Stride Observability

The tokenization operator observes the signal only at stride-separated locations. This induces an effective observation frequency

$$f_{\text{eff}} = \frac{f_s}{S}, \quad (16)$$

Signal dynamics occurring at temporal scales shorter than the stride interval become progressively harder to observe through the token sequence.

Under phase-locking conditions, consecutive tokens become identical and the local representation collapses into a stationary state. Fine-grained phase evolution and high-frequency variations occurring between token boundaries therefore become hidden from the observable representation presented to the model.

Analysis of Structural Degeneracies and Invariances

When $f \in \mathcal{F}_{\text{lock}}$, the resulting behavior depends on the relationship between P and S .

Case 1: Equal Patch Size and Stride ($P = S$) When patches are contiguous and non-overlapping,

$$P = S. \quad (17)$$

Equation (9) directly implies

$$\mathbf{t}_{j+1} = \mathbf{t}_j \quad \forall j. \quad (18)$$

The token sequence collapses into a static series of identical vectors, eliminating all observable temporal variation.

Case 2: Overlapping Windows ($S < P$) When the stride is smaller than the patch length,

$$S < P, \quad (19)$$

the translational invariance condition remains valid:

$$x[n + S] = x[n]. \quad (20)$$

Therefore,

$$\mathbf{t}_{j+1} = \mathbf{t}_j. \quad (21)$$

Even though each token spans a larger temporal context, exact stride-period alignment forces consecutive tokens to remain identical. Consequently, overlap alone cannot prevent token-space degeneracy.

Case 3: Disjoint Windows ($S > P$) When

$$S > P, \quad (22)$$

gaps of length $S - P$ appear between successive windows. Under the phase-locking condition, which still guarantees

$$\mathbf{t}_{j+1} = \mathbf{t}_j. \quad (23)$$

However, the interval separating consecutive windows remains completely unobserved by the tokenization operator. Consequently, distinct signals may produce exactly the same token sequence despite exhibiting different dynamics within the hidden intervals.

In this regime, the observation operator becomes non-injective: multiple underlying trajectories can map to an identical token representation. This creates a stronger form of information loss than simple token degeneracy because portions of the signal become permanently unrecoverable from the tokenized representation.

Domain Description and Ontology Representation

The considered domain takes into account the energy generated by an asset infrastructure centered around a photovoltaic array (`pv:PhotovoltaicArray`), operating within a localized smart micro-grid ecosystem. This environment is dynamically monitored, analyzed, and managed by an autonomous AI control agent (`pv:ControlAgent`) that balances power flows and detects system failures.

Physical Assets and Grid Topology

The physical core of the system is composed of four primary entities under the physical component (`pv:PhysicalComponent`) class, which are physically connected to (`pv:connectedTo`) one another to form the micro-grid's electrical topology:

- The photovoltaic array serves as the primary generation source, defined by its physical size (`pv:panelCount`) and inverter nameplate capacity (`pv:nominalPowerW`).
- The array directly supplies power to (`pv:suppliesTo`) three downstream assets: the external national electricity grid (`pv:NationalGrid`) interface (an infinite bidirectional bus for energy exchange), local internal consumption loads (`pv:InternalLoad`) representing dynamic demand, and a local electrochemical storage unit (`pv:BatteryStorage`).
- Functionally, the battery storage unit actively buffers (`pv:buffers`) the photovoltaic array, absorbing excess generation or discharging power during deficits.

Telemetry Streams and Tokenization Layer

To evaluate system performance, data streams are continuously collected across two operational vectors:

- The photovoltaic array measures (`pv:measures`) a continuous time-series signal (`pv:Signal`). This stream encapsulates instantaneous data properties such as photovoltaic power (`pv:pvPowerW`), global horizontal irradiance (`pv:horizontalIrradianceWm2`), and tilted surface irradiance (`pv:tiltedIrradianceWm2`), alongside frequency-domain metadata including the raw sampling rate (`pv:samplingFrequencyHz`), dominant periodic cycle frequency (`pv:fundamentalFrequencyHz`), wave amplitude (`pv:amplitudeW`), and phase offset relative to epoch (`pv:phaseRad`).
- Concurrently, the array is mapped via an environmental snapshot relationship (`pv:hasWeatherObservation`) to a weather observation log (`pv:WeatherObservation`), which tracks atmospheric variables like ambient air temperature (`pv:airTemperatureC`), wind speed (`pv:windSpeedMs`), and wind direction (`pv:windDirectionDeg`).

To prepare this continuous telemetry for deep learning inference, the raw signal passes through a processing layer that tokenizes (`pv:tokenizes`) it into a patched signal representation (`pv:PatchedSignal`). This sequence chunking strategy, tailored for foundation architectures like Chronos, is explicitly governed by architectural attributes: the token block size (`pv:patchSize`), window step stride (`pv:stride`), and calculated overlap ratio (`pv:overlapRatio`).

Inferred Environmental Contexts and System Anomalies

An execution reasoning engine dynamically updates the operational metadata of the system based on physical and mathematical constraints. The system extracts a baseline generation state (`pv:GenerationState`) derived from the tilted surface irradiance feature, classifying the current solar environment into one of four mutually exclusive subclasses: full peak production clear sky (`pv:ClearSkyState`), intermittent cloud cover (`pv:PartialCloudState`), dense cloud cover overcast (`pv:OvercastState`), or zero usable irradiance nighttime (`pv:NighttimeState`).

Simultaneously, the reasoner assesses system health by validating whether the photovoltaic array exhibits an anomaly (`pv:hasAnomaly`). It detects six distinct anomaly states (`pv:AnomalyState`):

- **Clipping Anomaly (`pv:ClippingAnomaly`):** Occurs when solar power saturates at or above inverter nominal capacity during periods of high peak irradiance.
- **Inverter Trip Anomaly (`pv:InverterTripAnomaly`):** A hard breaker or component disconnection where output drops to zero despite significant solar resource availability.
- **Ramp Anomaly (`pv:RampAnomaly`):** Identified when the dynamic rate of power change (`pv:deltaPowerW`) between consecutive intervals exceeds safe structural ramp thresholds.
- **Soiling Anomaly (`pv:SoilingAnomaly`):** A persistent drop in normalized conversion efficiency, indicating panel surface contamination such as dust or soil accumulation.
- **Overheating Anomaly (`pv:OverheatingAnomaly`):** High ambient heat conditions inducing thermal derating, which depresses panel production efficiency under strong irradiance.
- **Malfunction Anomaly (`pv:MalfunctionAnomaly`):** A structural hardware fault where generation drops to a severely degraded, near-zero baseline without clear environmental causes.

Machine Learning Vulnerabilities: Aliasing Events

In parallel with physical faults, the semantic layer evaluates computational data-processing vulnerabilities. If tokenization window choices conflict with the signal’s cyclical boundaries, the patched signal exhibits aliasing (`pv:exhibitsAliasing`), indicating an ongoing aliasing event (`pv:AliasingEvent`) that compromises downstream forecasting model precision:

- A frequency blind spot (`pv:FrequencyBlindSpot`) is flagged when the continuous signal component matches an integer harmonic of the patch-induced frequency (`pv:patchInducedFrequencyHz`), rendering major cyclical patterns structurally invisible to the model (Hypotheses H1 and H1b).
- A phase aliasing effect (`pv:PhaseAliasingEffect`, Hypothesis H2) occurs when an active phase offset or spatial shift (`pv:phaseOffsetRad`) misaligns with token boundaries, degrading representation fidelity.
- An overlap distortion effect (`pv:OverlapEffect`, Hypothesis H3) reveals configuration setups where a low overlap ratio indicates suboptimal stride lengths, generating context sequence block redundancy.

Closed-Loop Autonomous Action

To restore grid equilibrium and preserve physical integrity, the photovoltaic array triggers actions (`pv:triggersAction`) belonging to the agent action (`pv:AgentAction`) class hierarchy. The autonomous control agent manages the closed-loop optimization loop and performs (`pv:performs`) four specialized mitigation interventions:

- It coordinates a battery dispatch intervention (`pv:BatteryDispatchAction`) to buffer grid fluctuations under volatile weather or to substitute lost generation during unexpected inverter breaker trips.
- It deploys a generation curtailment action (`pv:CurtailGenerationAction`) to throttle active power spikes during inverter clipping or critical panel overheating states.
- It initiates demand side load adjustment changes (`pv:LoadAdjustmentAction`) to shift internal consumer consumption steps during low overcast or zero production nighttime states.
- It issues a rapid warning alert (`pv:RampAlertAction`) to stability systems the instant an active ramp rate anomaly is detected.

Structural Aliasing Expected Impact

To ground the experimental validation within the semantic framework established by the system ontology, we map the digital signal processing phenomena directly to the ontological layer. Let a continuous-time

physical signal (such as `pv:pvPowerW` or `pv:tiltedIrradianceWm2`) be sampled uniformly at a rate f_s (`pv:samplingFrequencyHz`) and transformed into a tokenized sequence via a `pv:PatchedSignal` operator defined by patch size P (`pv:patchSize`) and translation stride S (`pv:stride`).

The architectural grid frequency f_p (`pv:patchInducedFrequencyHz`) governs the macro-temporal resolution of the token space and is defined as:

$$f_p = \frac{f_s}{S} \quad (24)$$

Based on this parametric environment, we define three distinct, falsifiable hypotheses to characterize the boundaries of structural aliasing.

Hypothesis 1 (H_1): Frequency-Dependent Blind Spots

- **Ontological Alignment:** `pv:FrequencyBlindSpot` (asserted via SWRL inference rules R18 and R19).
- **Statement:** When the fundamental frequency f (`pv:fundamentalFrequencyHz`) of an input time-series signal matches an integer multiple of the patch-induced frequency f_p , the tokenization layer maps distinct cyclical variations into static, identical latent representations, creating discrete informational blind spots across the spectrum.
- **Mathematical Boundary:** An aliasing event is triggered when the frequency ratio satisfies the harmonic condition:

$$f = n \cdot f_p = n \left(\frac{f_s}{S} \right), \quad n \in \mathbb{N}^+ \quad (25)$$

Under this boundary condition, the column rank of the local trajectory matrix collapses to unity, rendering the internal dynamics of the patch unobservable to downstream self-attention mechanisms.

- **Falsifiability Criterion:** This hypothesis is falsified if empirical probing models or closed-loop reconstruction tasks demonstrate uniform representation fidelity across the operational band $[2, \frac{f_s}{2}]$ Hz, without exhibiting statistically significant localized performance degradation or info-loss anomalies at the exact integer harmonics of f_p .

Hypothesis 2 (H_2): Phase Alignment and Temporal Displacement

- **Ontological Alignment:** `pv:PhaseAliasingEffect` (asserted via SWRL inference rule R20).
- **Statement:** The magnitude of structural information loss at an aliased harmonic frequency is non-deterministic and varies as a function of the temporal alignment between the signal’s peak-trough dynamics and the rigid boundaries of the tokenization grid.
- **Mathematical Boundary:** For a fixed harmonic frequency $f \in \mathcal{F}_{\text{lock}}$, translating the initial phase offset ϕ (`pv:phaseRad`) within the discrete-time sinusoid:

$$x[n] = \sin \left(2\pi \frac{f}{f_s} n + \phi \right), \quad \phi \in [0, 2\pi) \quad (26)$$

modifies the specific sub-sequence frozen inside each individual patch window. If the phase shift satisfies $\phi \in (0, S)$, it induces a structural misalignment that alters the component-wise values of consecutive token vectors, systematically modulating the token space fidelity.

- **Falsifiability Criterion:** This hypothesis is falsified if information-theoretic tracking metrics (such as the Space Saving metric SV) or empirical probing classification accuracies remain strictly invariant across all randomly generated phase displacements $\phi \sim \text{Uniform}(0, 2\pi)$ at a locked harmonic frequency, proving that boundary location exercises no control over latent feature degradation.

Hypothesis 3 (H_3): Geometric Configuration and Overlap Mechanics

- **Ontological Alignment:** `pv:OverlapEffect` (asserted via SWRL inference rule R21).
- **Statement:** Altering the geometric relationship between the patch window length and the translational step size shifts the failure mode of the token space, moving the representation from absolute vector degeneracy to high-dimensional contextual redundancy.

- **Mathematical Boundary:** The geometric interaction is dictated by the token overlap ratio or (`pv:overlapRatio`):

$$or = \frac{P - S}{P} \quad (27)$$

When $or = 0$ ($P = S$), the windowing is strictly contiguous and non-overlapping; satisfying Equation (2) forces consecutive tokens to be mathematically identical ($\mathbf{t}_j = \mathbf{t}_{j+1}$), wiping out temporal gradients. When $or \rightarrow 1$ ($S \ll P$), successive patches share massive historical context; although the stride-bound phase-locking condition $x[n + S] = x[n]$ still holds, the expanded patch dimension P forces tokens to capture intra-patch phase variations that break the absolute vector identity, shifting the error into an over-smoothed informational redundancy.

- **Falsifiability Criterion:** This hypothesis is falsified if varying the architectural parameters P and S while keeping a locked signal frequency f yields an identical trajectory matrix rank profile and an identical distribution of downstream transformer attention weights, proving that the geometric layout of the patch grid does not modulate the structural aliasing expression.

Signal Generation

In order to evaluate the impact of structural aliasing on the Chronos-Bolt framework and its forecasting capabilities, we conducted a series of experiments using synthetic time-series data.

To observe the predicted values generated by Chronos-Bolt, we use three types of signal generators: a single-harmonic sinusoidal generator, a modified TSMixup generator, and a KernelSynth generator. The last two are based on methods proposed for Chronos training.[1]

Single-Harmonic Sinusoidal Generator This generator produces a simple sinusoidal signal with a single frequency component. The generated signal can be expressed mathematically as:

$$x(t) = A \cdot \sin(2\pi ft + \phi) \quad (28)$$

The continuous signal is then sampled at a specified rate to create a discrete time-series. The parameters of the generator, such as amplitude, frequency, and phase, can be adjusted to create different sinusoidal signals for testing purposes.

Light TSMixup Generator We propose a controlled probing variant of the TSMixup method described in the Chronos data augmentation procedure.[1] The original TSMixup algorithm samples subsequences from real training datasets, applies mean scaling to each sampled time series, draws mixing coefficients from a symmetric Dirichlet distribution, and returns their convex combination.

Light TSMixup preserves this stochastic mixing structure, but replaces the real dataset collection with a controlled pool of sinusoidal signals. Each base signal is defined by frequency, amplitude, phase, length, and sampling frequency. This simplification makes the generated signals easier to interpret when testing whether Chronos-Bolt exhibits frequency-dependent blind spots or phase-locking effects.

The parameters of the generator, such as the maximum number of sinusoidal components, Dirichlet concentration, target length, and sampling frequency, can be adjusted to create different synthetic signals for testing purposes. The pseudocode for the Light TSMixup generator is listed in Appendix B.

KernelSynth Generator Since TSMixup is able to produce only simple composition of sinusoidal signals, we implement KernelSynth as a more complex alternative, which can generate signals with richer frequency content and temporal dynamics. By comparing the performance of Chronos-Bolt on signals generated by both methods, we can gain insights into how structural aliasing affects the model’s ability to learn and forecast time-series data with varying levels of complexity.

The KernelSynth generator is also described in the Chronos data augmentation procedure.[1] Its pseudocode and kernel bank are listed in Appendix C.

This generator produces synthetic signals by combining different kernel functions, using randomly selected weights from a normal distribution. The selected kernels are combined using sum and product operators.

As parameter the generator takes the maximum number of kernels to combine, and the time series output length. The selected kernel functions, the pseudo-code of the method, and the parameters used are described in Appendix B.

Title

References

- [1] Abdul Fatir Ansari et al. *Chronos: Learning the Language of Time Series*. Nov. 4, 2024. DOI: 10.48550/arXiv.2403.07815. arXiv: 2403.07815 [cs.LG]. URL: <http://arxiv.org/abs/2403.07815> (visited on 05/13/2026). Pre-published.

Appendix A: Ontology Representation

The core ontology architecture bridges the gap between cyber-physical energy systems and deep learning signal processing domain knowledge. It categorizes the application setting into five main conceptual pillars rooted under the ontological hierarchy:

- **Physical Subsystem (pv:PhysicalComponent):** Represents the hardware infrastructure of the micro-grid, including the solar array, electrochemical storage, consumer loads, and the macro grid tie.
- **Cognitive Layer (pv:ControlAgent):** The artificial intelligence system responsible for autonomous grid optimization, load balancing, dispatch scheduling, and proactive vulnerability mitigation.
- **Signal Processing Layer (pv:Signal, pv:PatchedSignal):** Tracks the raw telemetry inputs (both time-domain features and frequency-domain components) along with their tokenized representation configurations (patch dimensions, stride values) tailored for the Chronos foundation architecture.
- **Environmental Tracking (pv:WeatherObservation):** Captures concurrent meteorological snapshots (air temperature, wind vectors) influencing structural physics and electrical efficiency.
- **Inferred Operational States:** Dynamic metadata classified into generation levels (pv:GenerationState), systemic disruptions (pv:AnomalyState), control directives (pv:AgentAction), and signal degradation hazards (pv:AliasingEvent).

Core Concept Taxonomy

The formal classes defined within the schema are explicitly detailed below:

- pv:PhotovoltaicArray** The primary renewable energy asset. Characterized by panel count and nameplate nominal capacity. Source of telemetry streams; associated with weather conditions and subject to environmental/electrical anomalies.
- pv:BatteryStorage** Local electrochemical reservoir capable of dynamic grid connection/isolation. Characterized by capacity and instantaneous State of Charge (SoC).
- pv:InternalLoad** Represents internal industrial or residential consumption demands. Subject to adjustable, demand-side management interventions.
- pv:NationalGrid** The external electrical macro-grid interface, acting as a bidirectional infinite bus for excess export or deficit imports.
- pv:Signal** Time-series measurement stream containing global horizontal irradiance (G_h), tilted surface irradiance (G_{tilt}), instantaneous power output (P_{pv}), and analytical frequency descriptors (sampling rate, fundamental frequency, amplitude, phase).
- pv:PatchedSignal** The Chronos-specific tokenized structure derived from a raw signal. Governed by a selected window chunking mechanism (*patchSize*, *stride*), establishing a structural *overlapRatio* and a downstream *patchInducedFrequency*.

Semantic Relations & Properties

Object Properties (Entity-to-Entity Relationships)

The physical, functional, and inferred linkages across concepts are formalized using the following object properties:

- $\langle \text{pv:PhotovoltaicArray} \rangle \text{ pv:measures } \langle \text{pv:Signal} \rangle$
- $\langle \text{pv:PhotovoltaicArray} \rangle \text{ pv:hasWeatherObservation } \langle \text{pv:WeatherObservation} \rangle$
- $\langle \text{pv:PhotovoltaicArray} \rangle \text{ pv:hasGenerationState } \langle \text{pv:GenerationState} \rangle$ (*Inferred via SWRL*)
- $\langle \text{pv:PhotovoltaicArray} \rangle \text{ pv:hasAnomaly } \langle \text{pv:AnomalyState} \rangle$ (*Inferred via SWRL*)
- $\langle \text{pv:PhotovoltaicArray} \rangle \text{ pv:triggersAction } \langle \text{pv:AgentAction} \rangle$ (*Inferred via SWRL*)

- $\langle \text{pv:Signal} \rangle \text{ pv:tokenizes } \langle \text{pv:PatchedSignal} \rangle$
- $\langle \text{pv:PatchedSignal} \rangle \text{ pv:exhibitsAliasing } \langle \text{pv:AliasingEvent} \rangle$ (*Inferred via SWRL*)
- $\langle \text{pv:BatteryStorage} \rangle \text{ pv:buffers } \langle \text{pv:PhotovoltaicArray} \rangle$
- $\langle \text{pv:PhysicalComponent} \rangle \text{ pv:connectedTo } \langle \text{pv:PhysicalComponent} \rangle$ (*Symmetric Property*)

Data Properties (Entity Attributes)

Key quantitative attributes defining system conditions include:

- **Electrical/Solar:** $\text{pv:pvPowerW } (P_{\text{pv}})$, $\text{pv:nominalPowerW } (P_{\text{nom}})$, $\text{pv:tiltedIrradianceWm2 } (G_{\text{tilt}})$, $\text{pv:deltaPowerW } (\Delta P)$.
- **Meteorological:** $\text{pv:airTemperatureC } (T_{\text{air}})$, $\text{pv:windSpeedMs } (W_s)$.
- **Tokenization Architecture:** $\text{pv:patchSize } (N_p)$, $\text{pv:stride } (S)$, $\text{pv:overlapRatio } (\gamma)$, $\text{pv:patchInducedFrequency } (f_p)$.
- **Signal Descriptors:** $\text{pv:samplingFrequencyHz } (f_s)$, $\text{pv:fundamentalFrequencyHz } (f_0)$, $\text{pv:phaseRad } (\phi)$.

Knowledge-Driven Inference System (SWRL Rules)

Real-time diagnostic, classification, and mitigation logic are automated using a rule-based engine. The standard logical form of these production rules is detailed below:

Environmental Context Extraction

$$\begin{aligned} &\text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g \geq 800.0) \\ &\quad \longrightarrow \text{hasGenerationState}(?pv, \text{ClearSky}) \end{aligned} \quad (29)$$

$$\begin{aligned} &\text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \\ &\quad \wedge (?g \geq 150.0) \wedge (?g < 800.0) \longrightarrow \text{hasGenerationState}(?pv, \text{PartialCloud}) \end{aligned} \quad (30)$$

$$\begin{aligned} &\text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \\ &\quad \wedge (?g > 0.0) \wedge (?g < 150.0) \longrightarrow \text{hasGenerationState}(?pv, \text{Overcast}) \end{aligned} \quad (31)$$

$$\begin{aligned} &\text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g \leq 0.0) \\ &\quad \longrightarrow \text{hasGenerationState}(?pv, \text{Nighttime}) \end{aligned} \quad (32)$$

Cyber-Physical Anomaly Diagnostics

- **Inverter Clipping Anomaly (R5):**

$$\begin{aligned} &\text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{pvPowerW}(?s, ?p) \wedge \text{nominalPowerW}(?pv, ?nom) \\ &\quad \wedge (?p \geq ?nom) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 800.0) \\ &\quad \longrightarrow \text{hasAnomaly}(?pv, \text{Clipping}) \end{aligned} \quad (33)$$

- **Inverter Trip Anomaly (R6):**

$$\begin{aligned} &\text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{pvPowerW}(?s, ?p) \wedge (?p \leq 0.0) \\ &\quad \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 200.0) \longrightarrow \text{hasAnomaly}(?pv, \text{InverterTrip}) \end{aligned} \quad (34)$$

- **Ramp Event Anomaly (R7):**

$$\text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{deltaPowerW}(?s, ?dp) \wedge (?dp > 500.0) \longrightarrow \text{hasAnomaly}(?pv, \text{Ramp}) \quad (35)$$

- **Soiling Performance Deficit Anomaly (R8):**

$$\begin{aligned} & \text{PhotovoltaicArray}(?pv) \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 400.0) \\ & \wedge \text{pvPowerW}(?s, ?p) \wedge (?p > 100.0) \wedge \text{swrlb:divide}(?r, ?p, ?g) \wedge (?r < 0.3) \\ & \longrightarrow \text{hasAnomaly}(?pv, \text{Soiling}) \end{aligned} \quad (36)$$

- **Thermal Overheating Derating Anomaly (R9):**

$$\begin{aligned} & \text{PhotovoltaicArray}(?pv) \wedge \text{hasWeatherObservation}(?pv, ?w) \wedge \text{airTemperatureC}(?w, ?t) \wedge (?t > 35.0) \\ & \wedge \text{measures}(?pv, ?s) \wedge \text{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 600.0) \\ & \wedge \text{pvPowerW}(?s, ?p) \wedge \text{swrlb:divide}(?r, ?p, ?g) \wedge (?r < 0.4) \\ & \longrightarrow \text{hasAnomaly}(?pv, \text{Overheating}) \end{aligned} \quad (37)$$

Autonomous Closed-Loop Control Actuation

Based on inferred operating profiles, actions are dynamically mapped to handle grid stresses:

$$\text{pv:hasAnomaly}(?pv, \text{pv:Ramp}) \longrightarrow \text{pv:triggersAction}(?pv, \text{pv:RampAlert}) \quad (38)$$

$$\text{pv:hasAnomaly}(?pv, \text{pv:Clipping}) \vee \text{pv:hasAnomaly}(?pv, \text{pv:Overheating}) \quad (39)$$

$$\longrightarrow \text{pv:triggersAction}(?pv, \text{pv:CurtailGeneration}) \quad (40)$$

$$\text{pv:hasAnomaly}(?pv, \text{pv:InverterTrip}) \vee \text{pv:hasGenerationState}(?pv, \text{pv:PartialCloud}) \quad (41)$$

$$\longrightarrow \text{pv:triggersAction}(?pv, \text{pv:BatteryDispatch}) \quad (42)$$

$$\text{pv:hasGenerationState}(?pv, \text{pv:Overcast}) \vee \text{pv:hasGenerationState}(?pv, \text{pv:Nighttime}) \quad (43)$$

$$\longrightarrow \text{pv:triggersAction}(?pv, \text{pv:LoadAdjustment}) \quad (44)$$

Chronos Structural Aliasing Hypotheses

The key scientific contribution of this architecture lies in identifying information loss bounds during time-series patching. The model detects structural blind spots via frequency domain reasoning:

Hypothesis H1 & H1b: Patch-Induced Frequency Blind Spots

The patch tokenization process introduces an artificial operating frequency profile f_p , mathematically defined by the relationship:

$$f_p = \frac{f_s}{S} \quad (45)$$

Where f_s represents the raw Nyquist sampling rate and S denotes the token step stride.

- **Primary Blind Spot (H1):** If the continuous signal's structural fundamental frequency (f_0 , such as the daily diurnal solar cycle) aligns directly with the token boundaries, a tracking blind spot forms at

the fundamental harmonic region ($n = 1$):

$$\begin{aligned} & \text{Signal}(\text{?s}) \wedge \text{tokenizes}(\text{?s}, \text{?ps}) \wedge \text{fundamentalFrequencyHz}(\text{?s}, \text{?f}_0) \wedge \text{patchInducedFrequencyHz}(\text{?ps}, \text{?f}_p) \\ & \wedge \text{swrlb:divide}(\text{?r}, \text{?f}_0, \text{?f}_p) \wedge (\text{?r} \geq 0.9) \wedge (\text{?r} \leq 1.1) \\ & \longrightarrow \text{exhibitsAliasing}(\text{?ps}, \text{H1_BlindSpot}) \end{aligned} \quad (46)$$

- **Secondary Harmonic Blind Spot (H1b):** Re-occurs at the second harmonic ($n = 2$) boundary layer:

$$\begin{aligned} & \text{Signal}(\text{?s}) \wedge \text{tokenizes}(\text{?s}, \text{?ps}) \wedge \text{fundamentalFrequencyHz}(\text{?s}, \text{?f}_0) \wedge \text{patchInducedFrequencyHz}(\text{?ps}, \text{?f}_p) \\ & \wedge \text{swrlb:divide}(\text{?r}, \text{?f}_0, \text{?f}_p) \wedge (\text{?r} \geq 1.85) \wedge (\text{?r} \leq 2.15) \\ & \longrightarrow \text{exhibitsAliasing}(\text{?ps}, \text{H1b_BlindSpot}) \end{aligned} \quad (47)$$

Hypothesis H2: Phase Aliasing Effect

Vulnerabilities appear when temporal offsets misalign with chunk window anchors, corrupting early representation extraction features ($0 < \phi < S$):

$$\begin{aligned} & \text{Signal}(\text{?s}) \wedge \text{tokenizes}(\text{?s}, \text{?ps}) \wedge \text{phaseRad}(\text{?s}, \text{?}\phi) \wedge \text{stride}(\text{?ps}, \text{?S}) \\ & \wedge (\text{?}\phi > 0.0) \wedge (\text{?}\phi < \text{?S}) \longrightarrow \text{exhibitsAliasing}(\text{?ps}, \text{H2_Phase}) \end{aligned} \quad (48)$$

Hypothesis H3: Overlap Information Distortion

When the overlap configuration ratio ($\gamma = \frac{N_p - S}{N_p}$) drops below a critical information-theoretic bound ($\gamma < 0.5$), consecutive contextual patches display heavy text-token gaps, rendering trend forecasts highly unstable:

$$\text{PatchedSignal}(\text{?ps}) \wedge \text{overlapRatio}(\text{?ps}, \text{?or}) \wedge (\text{?or} < 0.5) \longrightarrow \text{exhibitsAliasing}(\text{?ps}, \text{H3_Overlap}) \quad (49)$$

Appendix B: Light TSMixup Pseudocode

Algorithm 1 Light TSMixup: Controlled Sinusoidal Time Series Mixup

Require: Sinusoidal pool $\mathcal{P} = \{p_1, \dots, p_{N_p}\}$, where each p_n is defined by frequency f_n , amplitude A_n , phase ϕ_n , source length T_n , and sampling frequency f_s ; maximum components $K = 10$; symmetric Dirichlet concentration $\alpha = 1.5$; minimum and maximum augmented lengths ($l_{\min} = 128, l_{\max} = 2048$).

Ensure: An augmented time series.

```

1:  $k \sim U\{1, K\}$                                 ▷ number of sinusoidal components to mix
2:  $l \sim U\{l_{\min}, l_{\max}\}$                         ▷ length of the augmented time series
3: for  $i \leftarrow 1$  to  $k$  do
4:    $n \sim U\{1, N_p\}$                                 ▷ sample a sinusoidal source
5:    $x_{1:l}^{(i)} \leftarrow \text{SampleSegment}(p_n, l, f_s)$     ▷ generate or crop a length- $l$  segment
6:    $\tilde{x}_{1:l}^{(i)} \leftarrow x_{1:l}^{(i)} / \left( \frac{1}{l} \sum_{j=1}^l |x_j^{(i)}| \right)$     ▷ apply mean scaling
7: end for
8:  $[\lambda_1, \dots, \lambda_k] \sim \text{Dir}([\alpha_1 = \alpha, \dots, \alpha_k = \alpha])$     ▷ sample mixing weights
9: return  $\sum_{i=1}^k \lambda_i \tilde{x}_{1:l}^{(i)}$                 ▷ take a weighted combination of scaled signals
```

Appendix C: KernelSynth Pseudocode and Kernel Bank

Algorithm 2 KernelSynth: Synthetic Data Generation using Gaussian Processes

Require: Kernel bank $\mathcal{K}$ listed in Table 1; maximum kernels per time series $J = 5$; generated length $l_{\text{syn}} = 1024$; optional numerical jitter $\epsilon = 0$.

Ensure: A synthetic time series $x_{1:l_{\text{syn}}}$.

```

1:  $j \sim U\{1, J\}$  ▷ sample the number of kernels
2:  $\{\kappa_1(t, t'), \dots, \kappa_j(t, t')\} \stackrel{\text{i.i.d.}}{\sim} \mathcal{K}$  ▷ sample kernels with replacement
3:  $t \leftarrow \text{linspace}(0, 1, l_{\text{syn}})$ 
4:  $\kappa^*(t, t') \leftarrow \kappa_1(t, t')$  ▷ initialize the composite covariance
5: for  $i \leftarrow 2$  to  $j$  do
6:    $\star \sim U\{+, \times\}$  ▷ sample a binary composition operator
7:    $\kappa^*(t, t') \leftarrow \kappa^*(t, t') \star \kappa_i(t, t')$  ▷ compose kernels
8: end for
9: if  $\epsilon > 0$  then
10:    $\kappa^*(t, t') \leftarrow \kappa^*(t, t') + \epsilon I$ 
11: end if
12:  $x_{1:l_{\text{syn}}} \sim \mathcal{GP}(0, \kappa^*(t, t'))$  ▷ sample from the GP prior
13: return  $x_{1:l_{\text{syn}}}$ 

```

Table 1: Kernel bank $\mathcal{K}$ used by KernelSynth.

Kernel	Formula	Hyperparameters
Constant	$\kappa_{\text{Const}}(x, x') = C$	$C = 1$
White Noise	$\kappa_{\text{White}}(x, x') = \sigma_n \cdot \mathbf{1}_{x=x'}$	$\sigma_n \in \{0.1, 1\}$
Linear	$\kappa_{\text{Lin}}(x, x') = \sigma_0^2 + x \cdot x'$	$\sigma_0 \in \{0, 1, 10\}$
RBF	$\kappa_{\text{RBF}}(x, x') = \exp\left(-\frac{\ x-x'\ ^2}{2l^2}\right)$	$l \in \{0.1, 1, 10\}$
Rational Quadratic	$\kappa_{\text{RQ}}(x, x') = \left(1 + \frac{\ x-x'\ ^2}{2\alpha}\right)^{-\alpha}$	$\alpha \in \{0.1, 1, 10\}, l = 1$
Periodic	$\kappa_{\text{Per}}(x, x') = \exp\left(-2 \sin^2\left(\pi \frac{\ x-x'\ }{p/l_{\text{syn}}}\right)\right)$	$p \in \{24, 48, 96, 168, 336, 672, 7, 14, 30, 60, 365, 730, 4, 26, 52, 4, 6, 12, 4, 40, 10\}$
